

Group Meeting

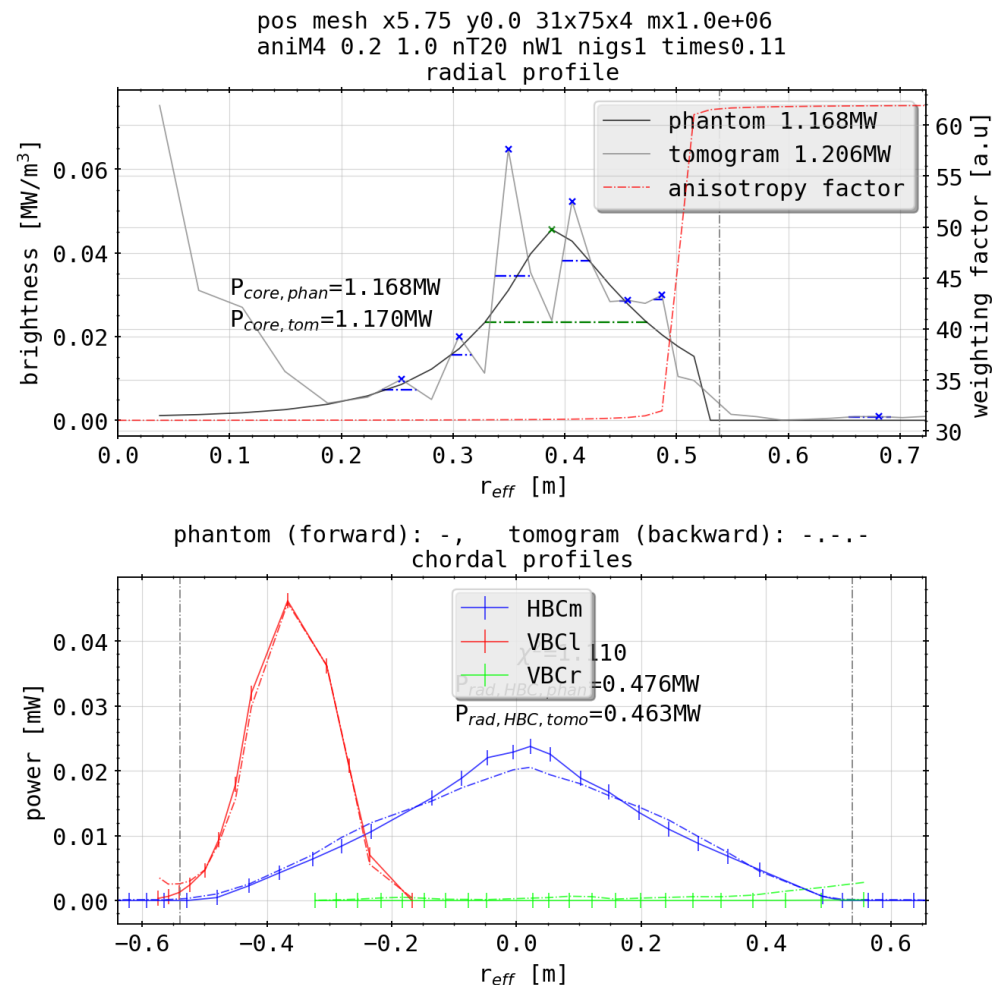
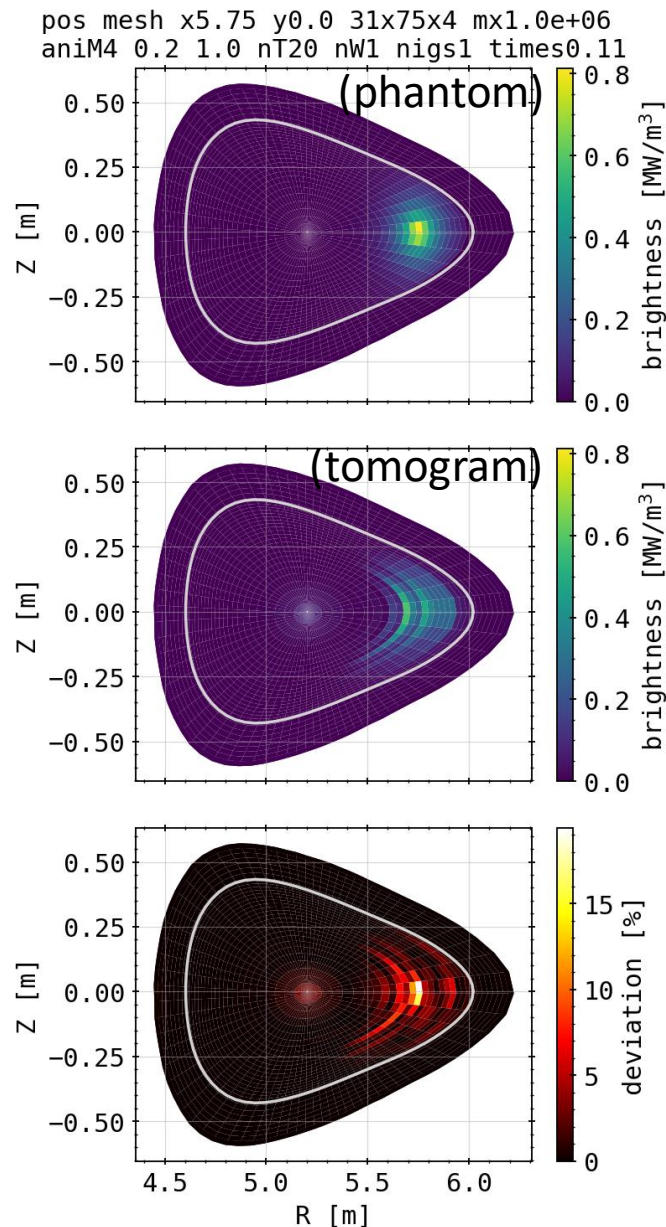
08/18/2020

HELMHOLTZ
SPITZENFORSCHUNG FÜR
GROSSE HERAUSFORDERUNGEN

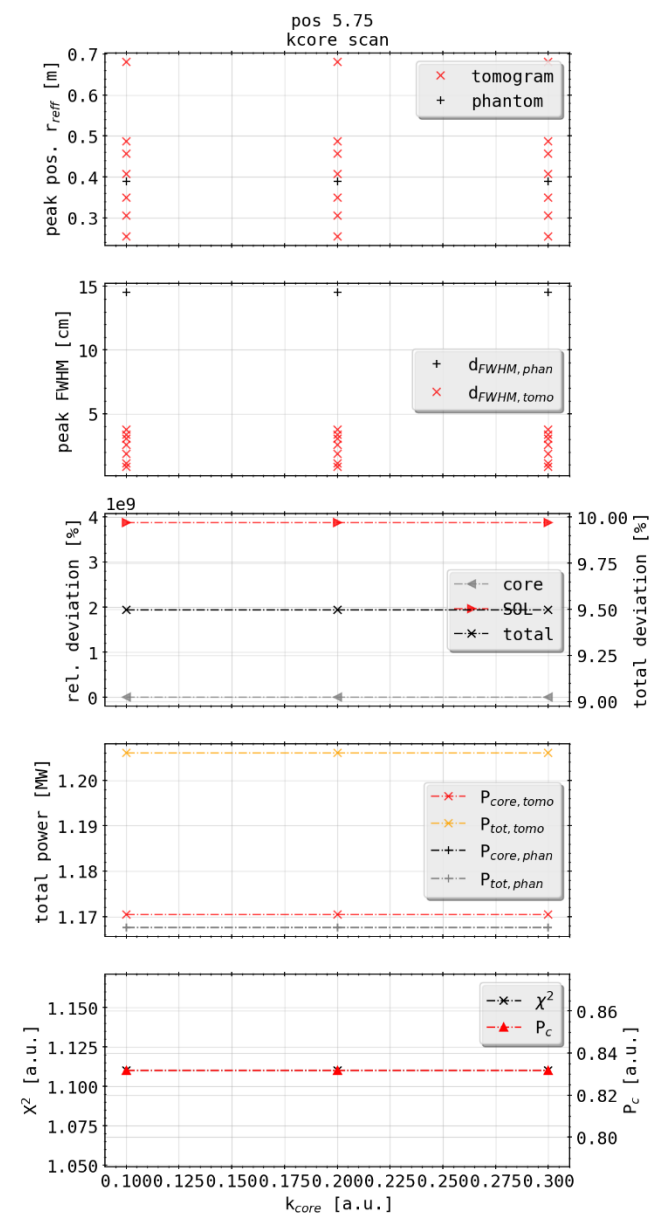


This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014-2018 and 2019-2020 under grant agreement No 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

Phantom Test: Single Spot Off-Center

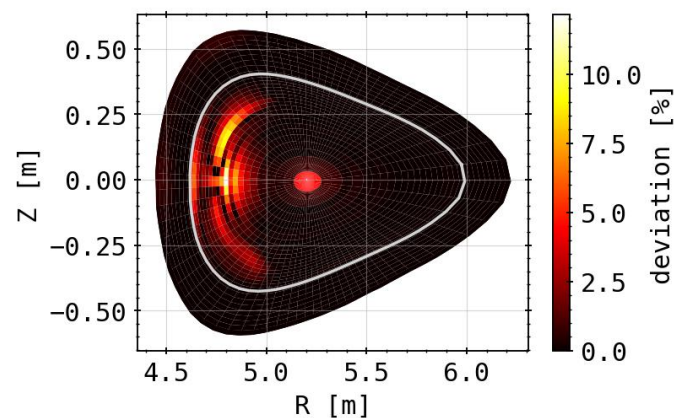
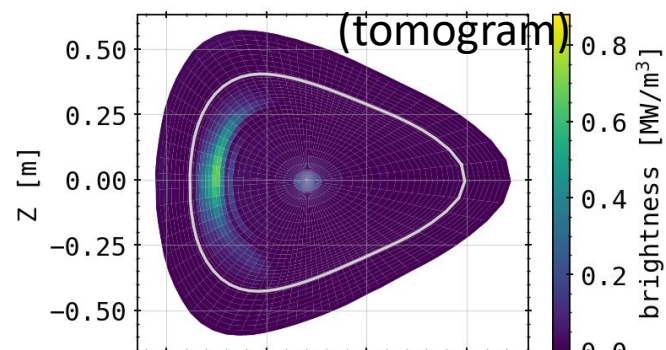
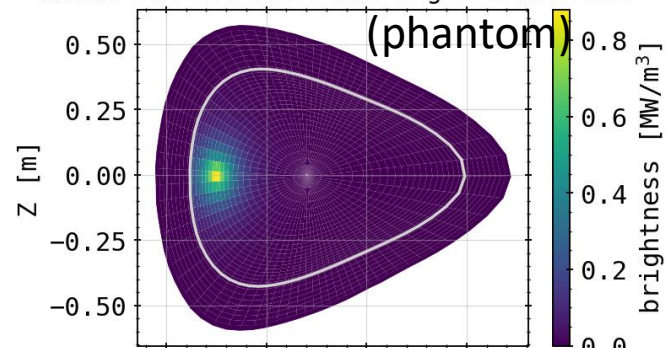


K scan at edge

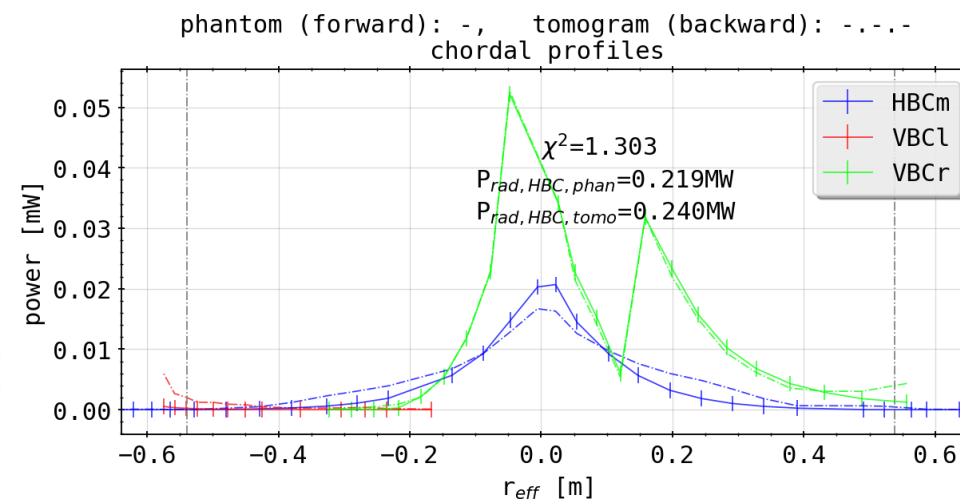
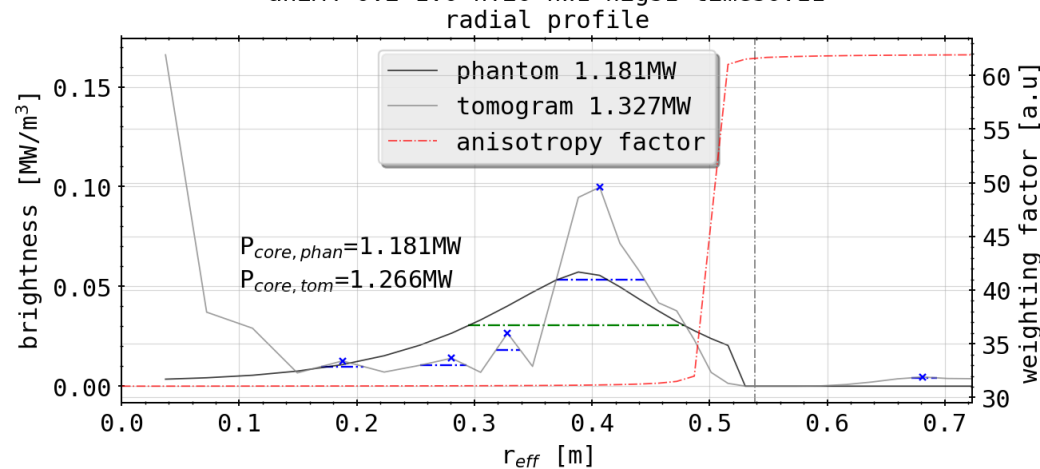


Phantom Test: Single Spot Off-Center

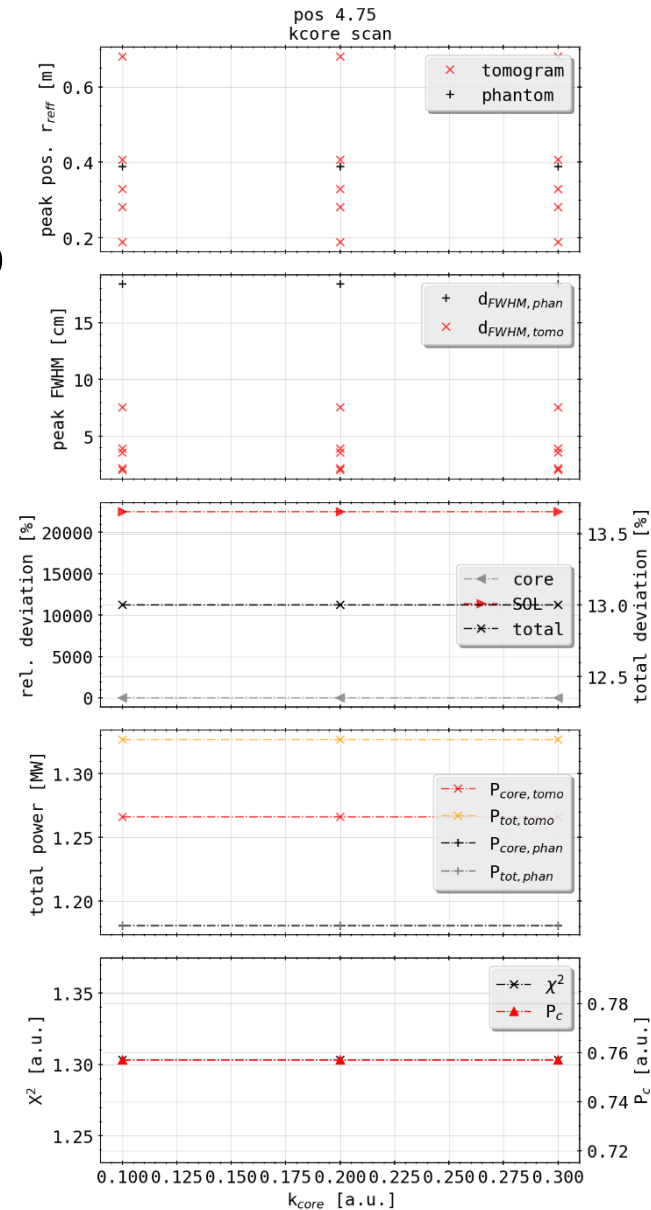
pos mesh x4.75 y0.0 31x75x4 mx1.0e+06
aniM4 0.2 1.0 nT20 nW1 nigs1 times0.11



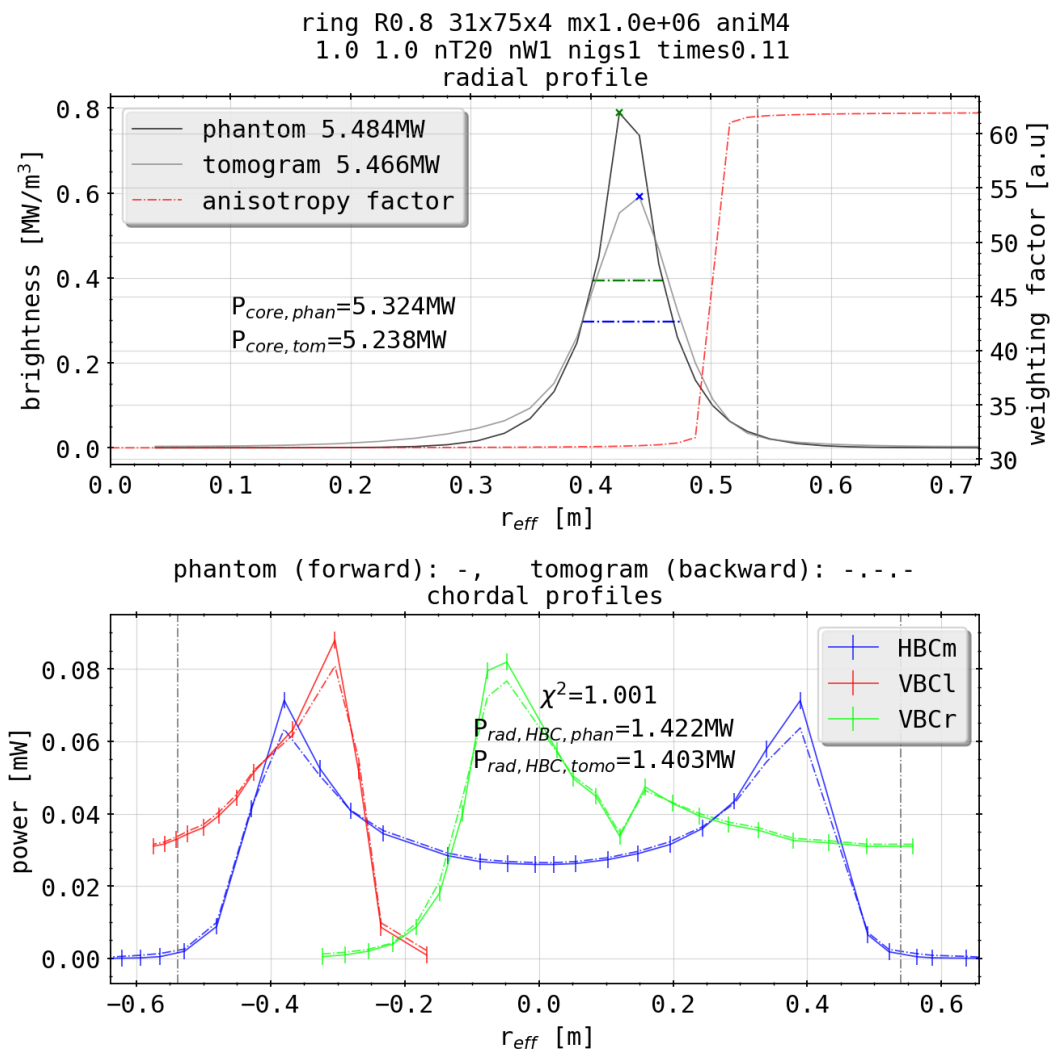
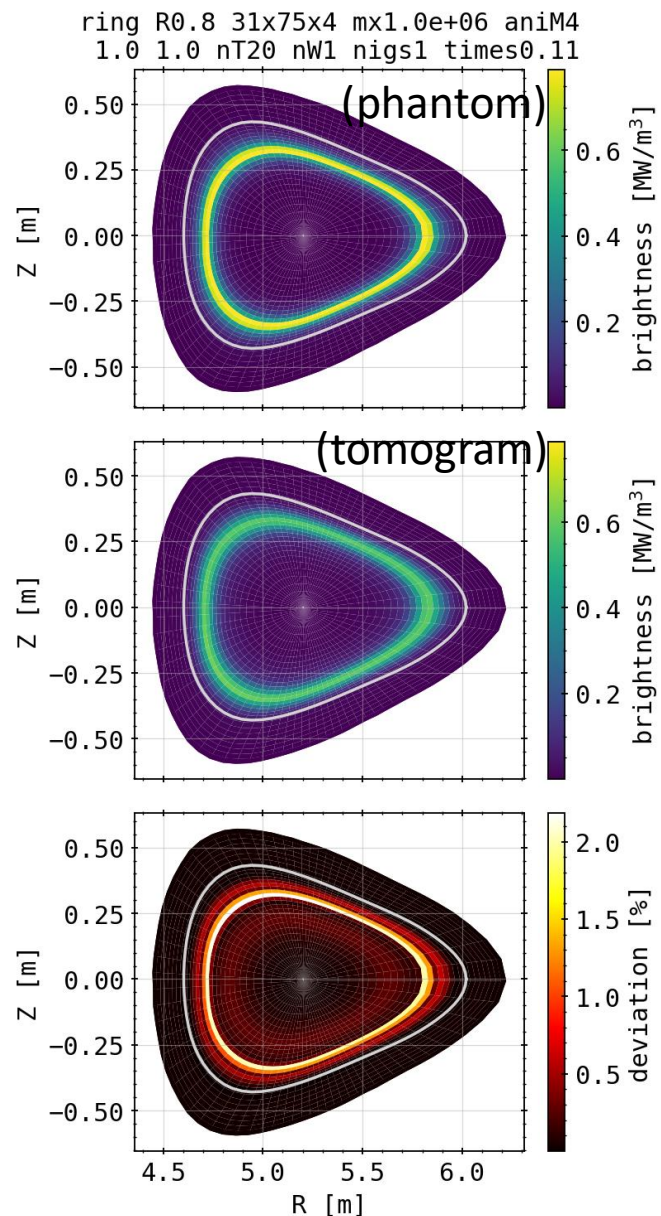
pos mesh x4.75 y0.0 31x75x4 mx1.0e+06
aniM4 0.2 1.0 nT20 nW1 nigs1 times0.11



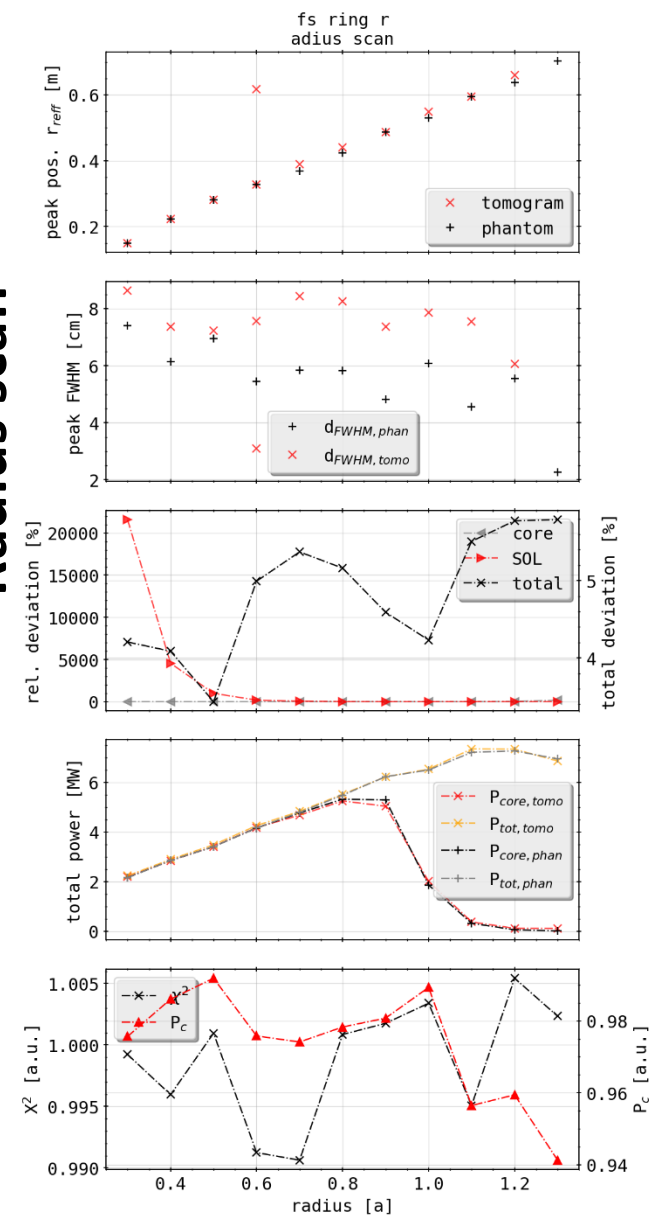
K scan at edge



Phantom Test: Bright Ring

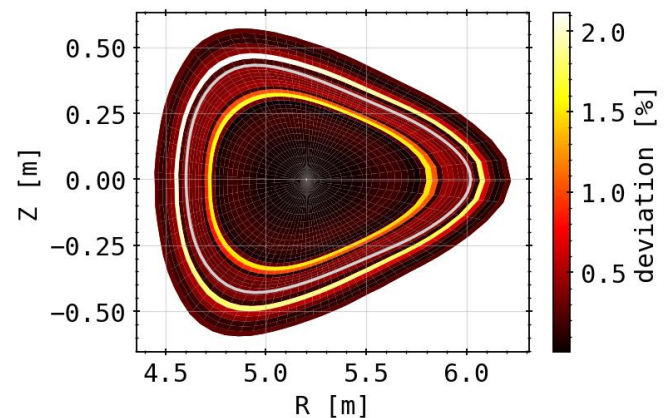
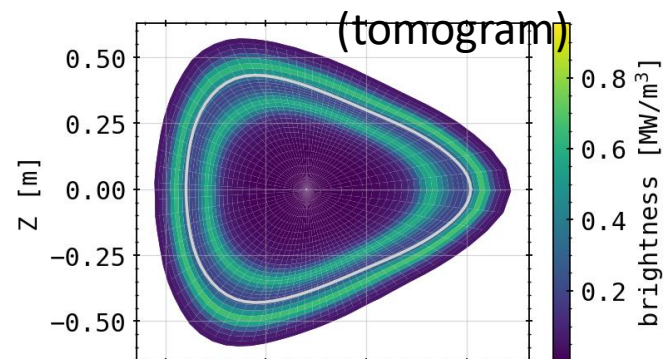
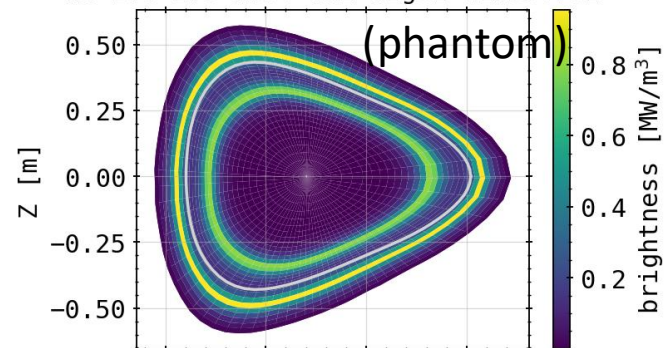


Radius scan

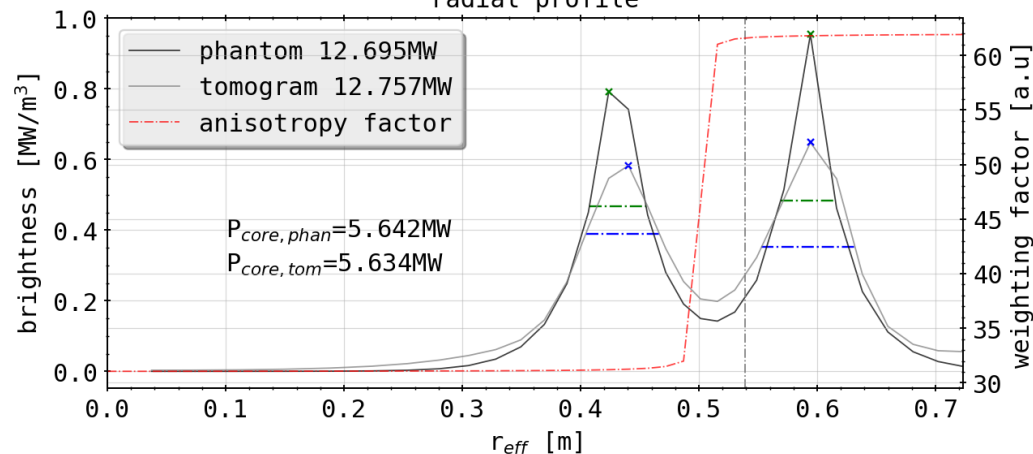


Phantom Test: Double Bright Ring

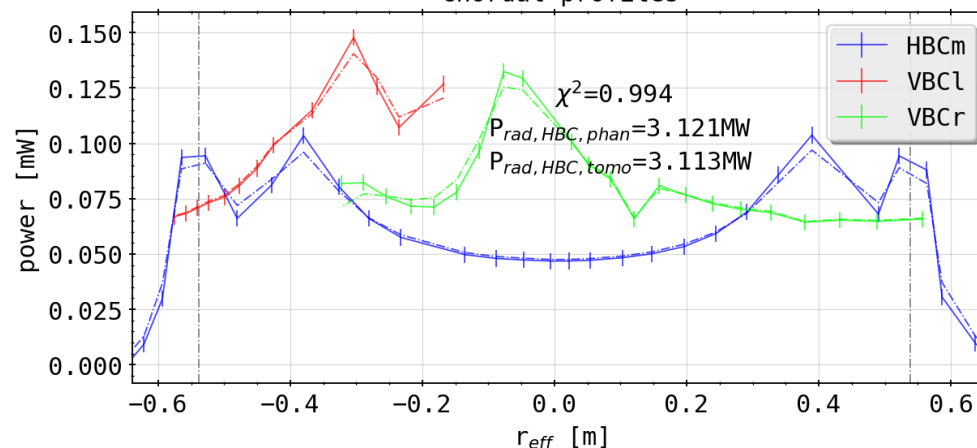
dfs R1.1 0.8 31x75x4 mx1.0e+06 ani
M4 1.0 1.0 nT20 nW1 nigs1 times0.11



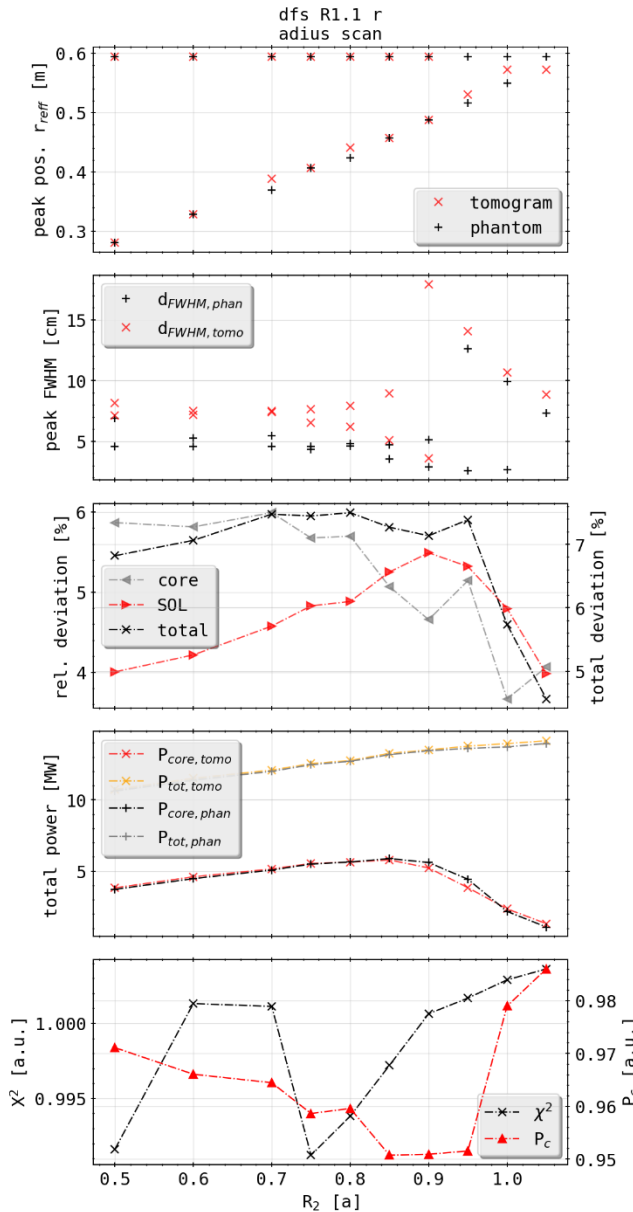
dfs R1.1 0.8 31x75x4 mx1.0e+06 ani
M4 1.0 1.0 nT20 nW1 nigs1 times0.11
radial profile



phantom (forward): -, tomogram (backward): -.-
chordal profiles

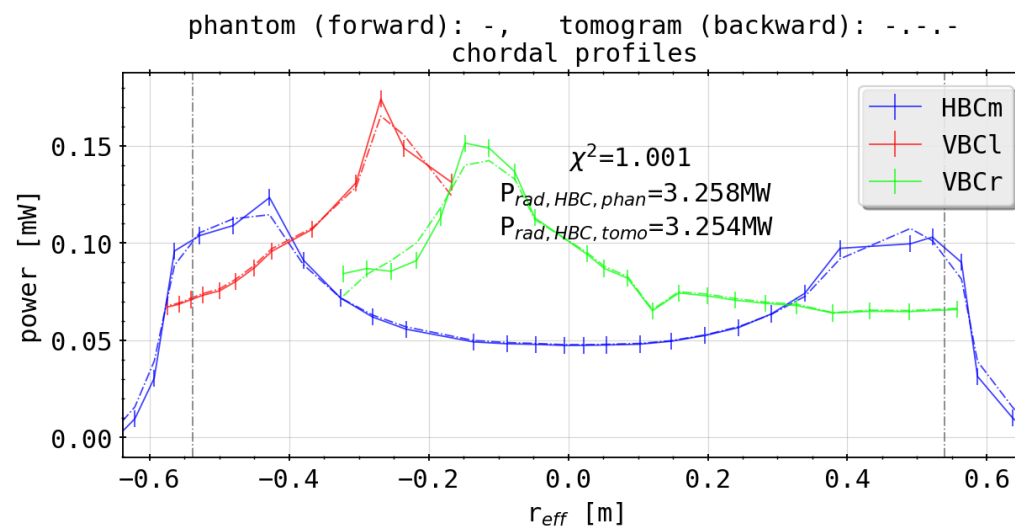
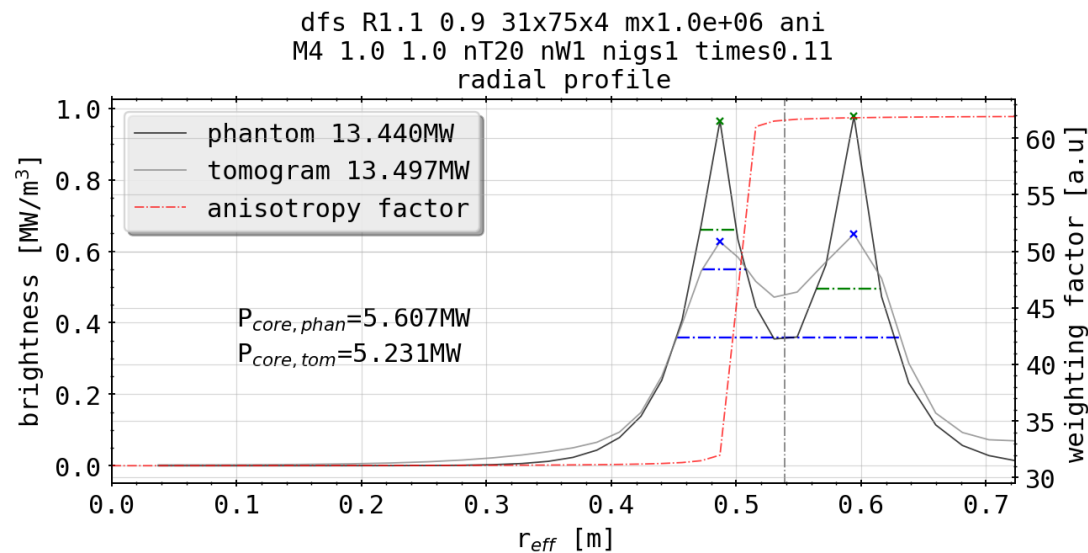
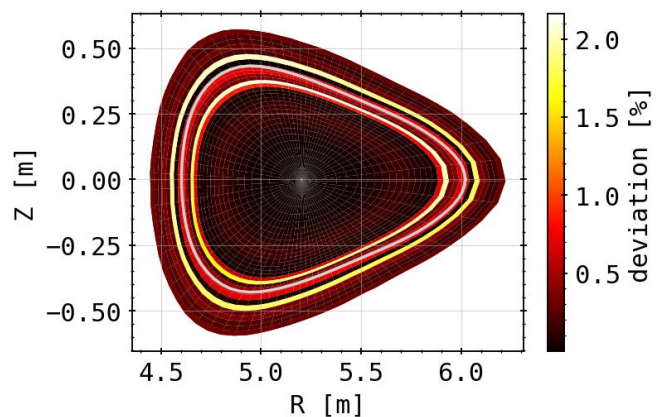
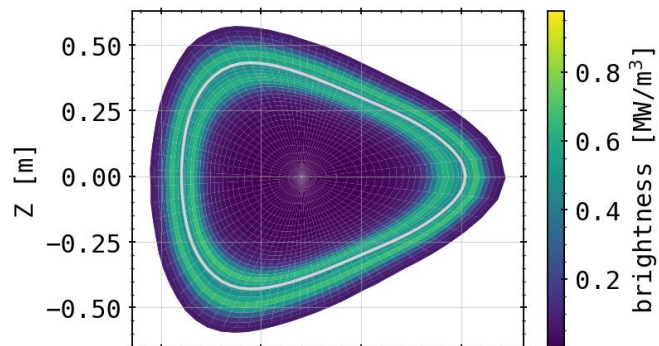
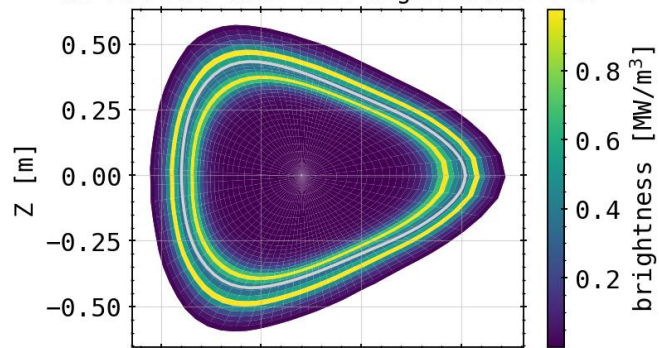


Radius scan

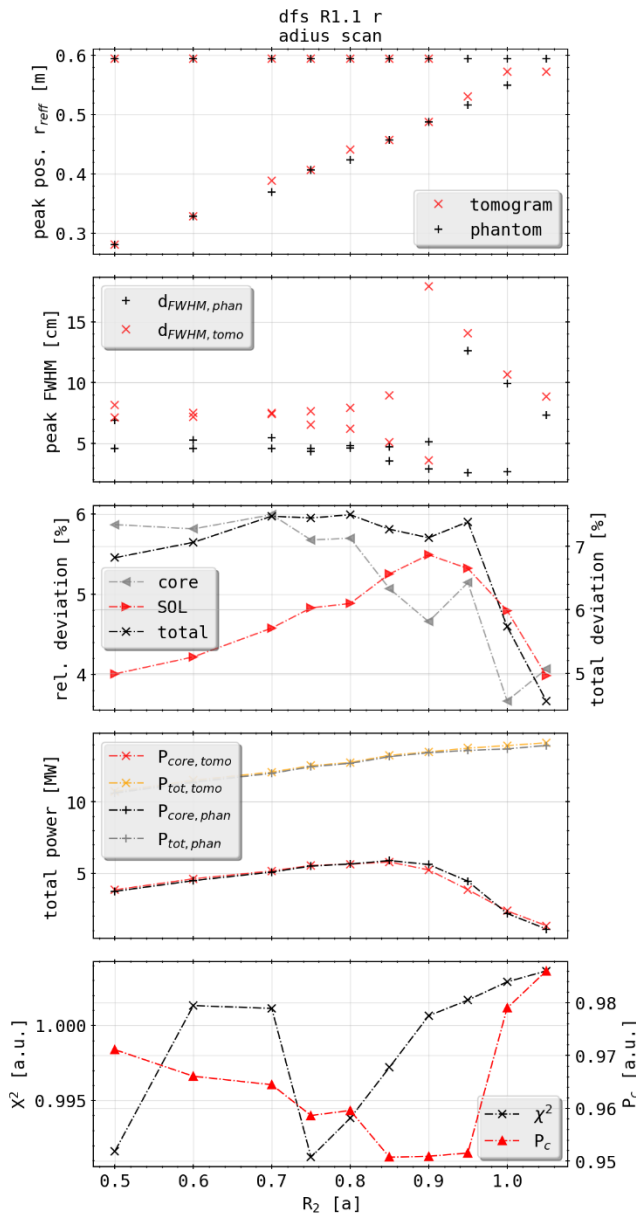


Phantom Test: Double Bright Ring

dfs R1.1 0.9 31x75x4 mx1.0e+06 ani
M4 1.0 1.0 nT20 nW1 nigs1 times0.11

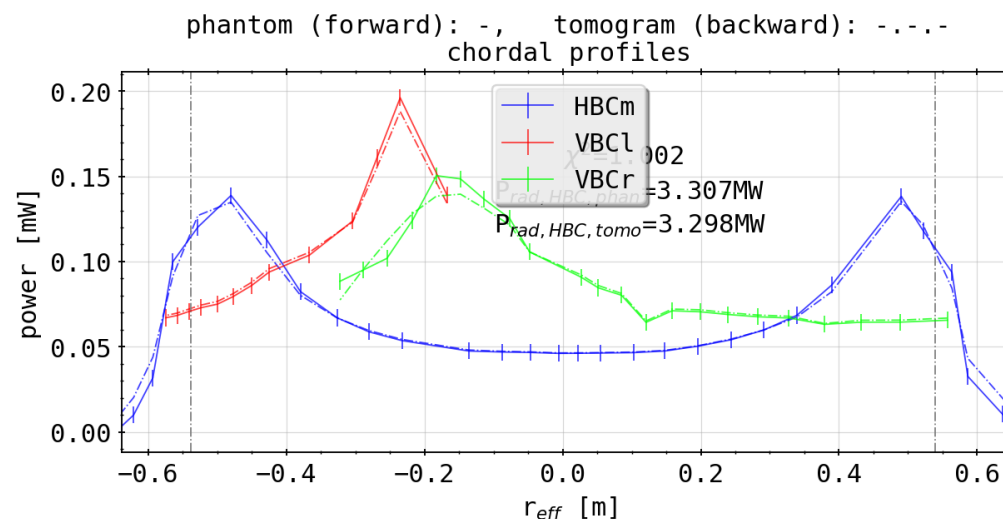
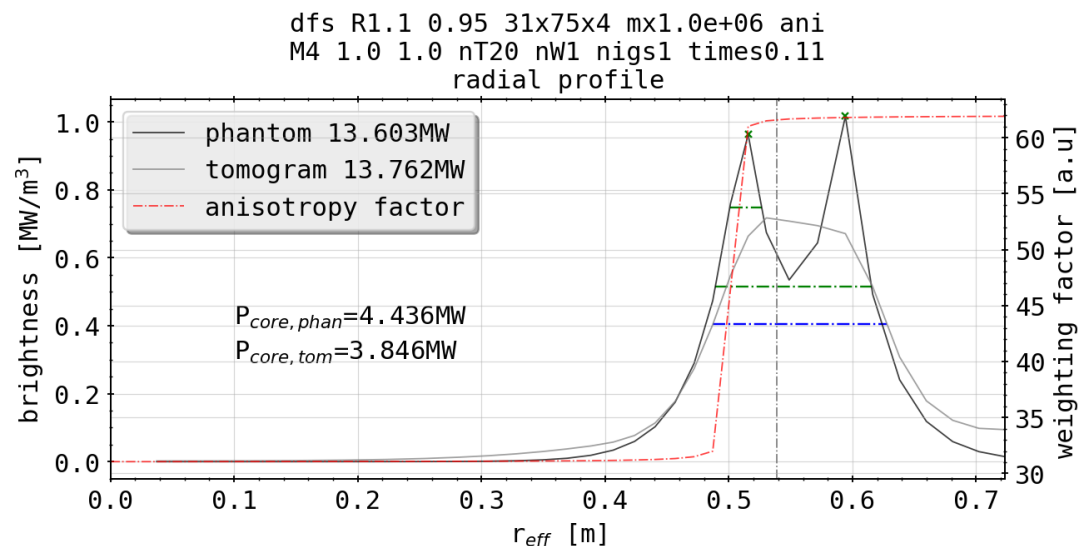
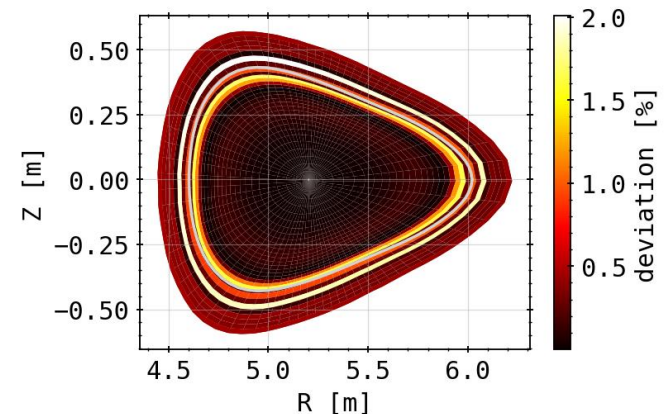
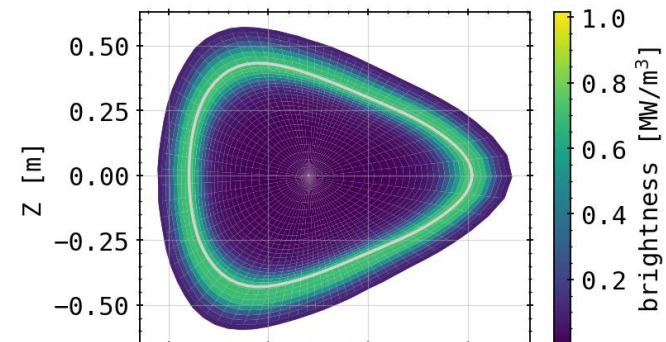
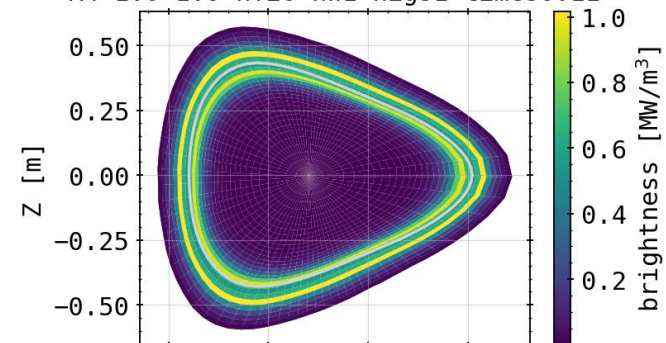


Radius scan

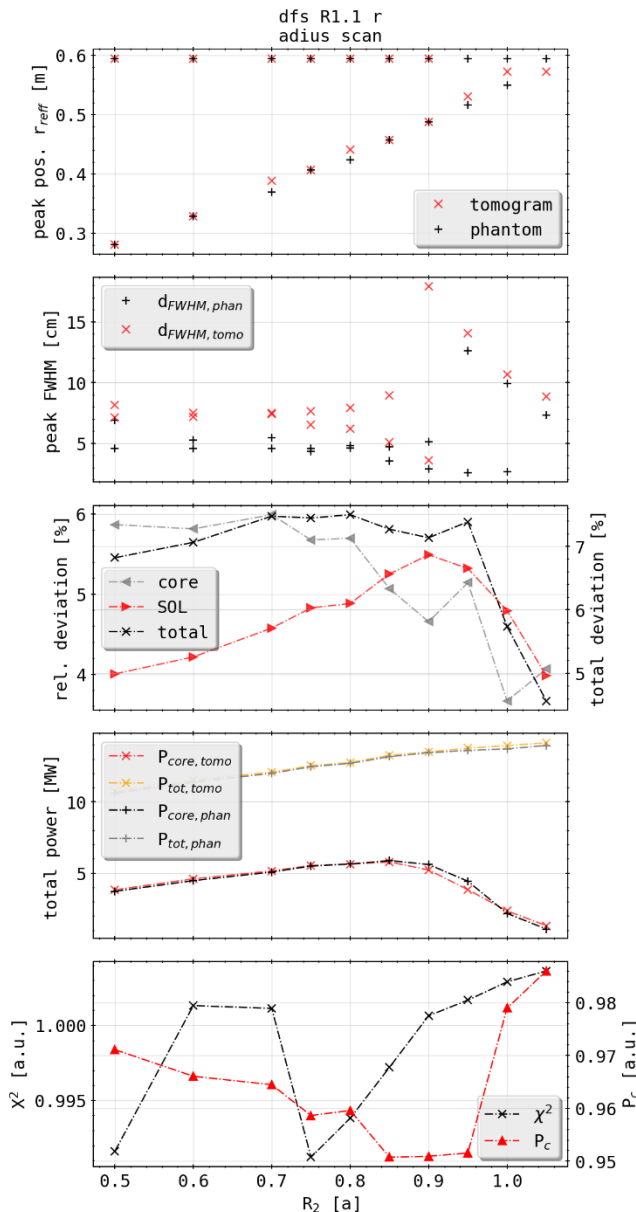


Phantom Test: Double Bright Ring

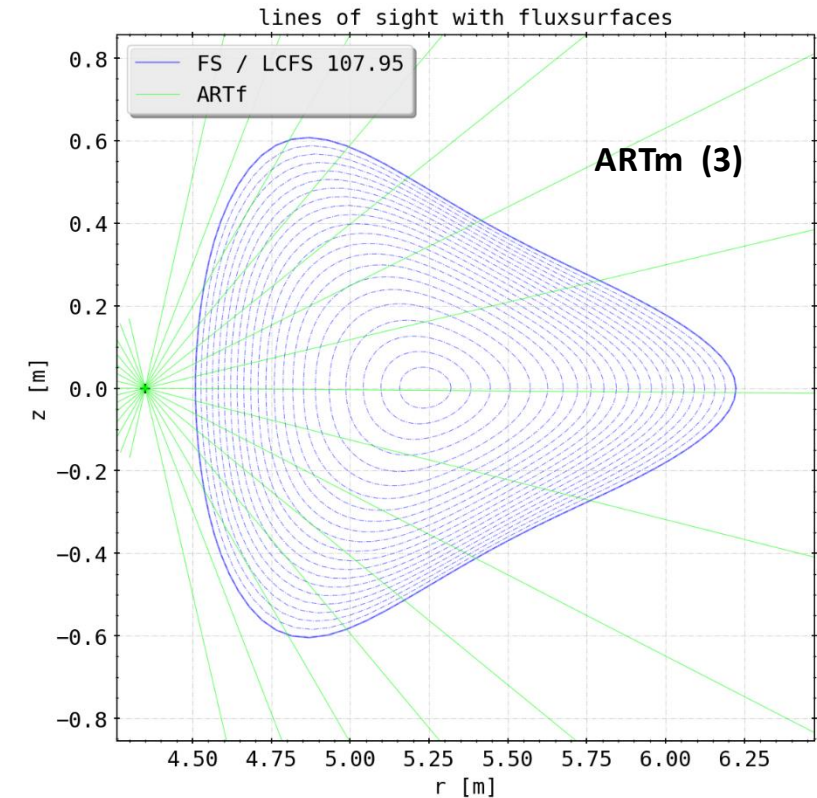
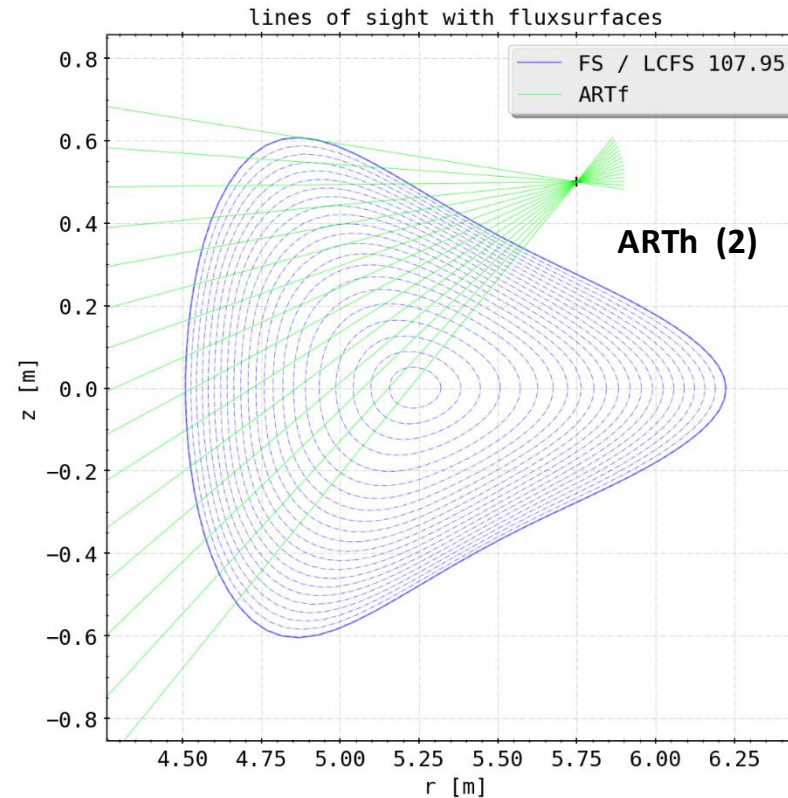
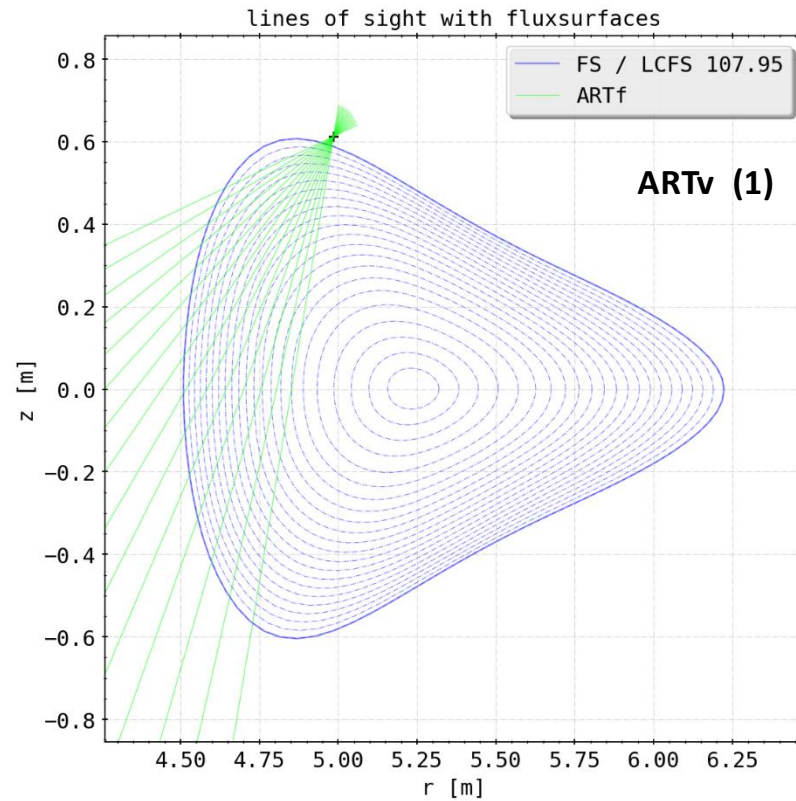
dfs R1.1 0.95 31x75x4 mx1.0e+06 ani
M4 1.0 1.0 nT20 nW1 nigs1 times0.11



Radius scan

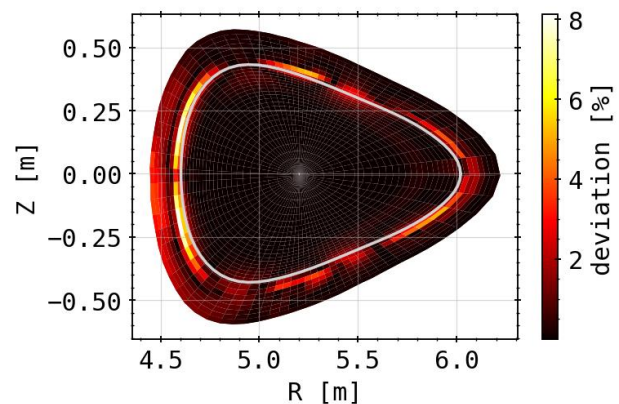
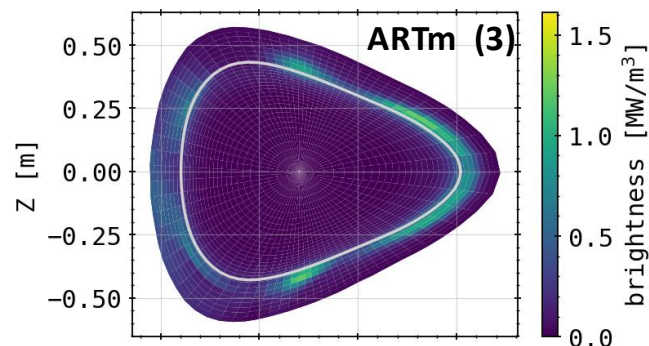
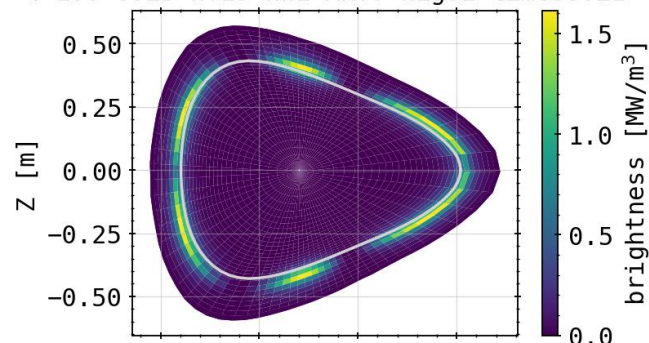


- adding new cameras at possible (1, 2) or impossible (3) positions, 15 channels each

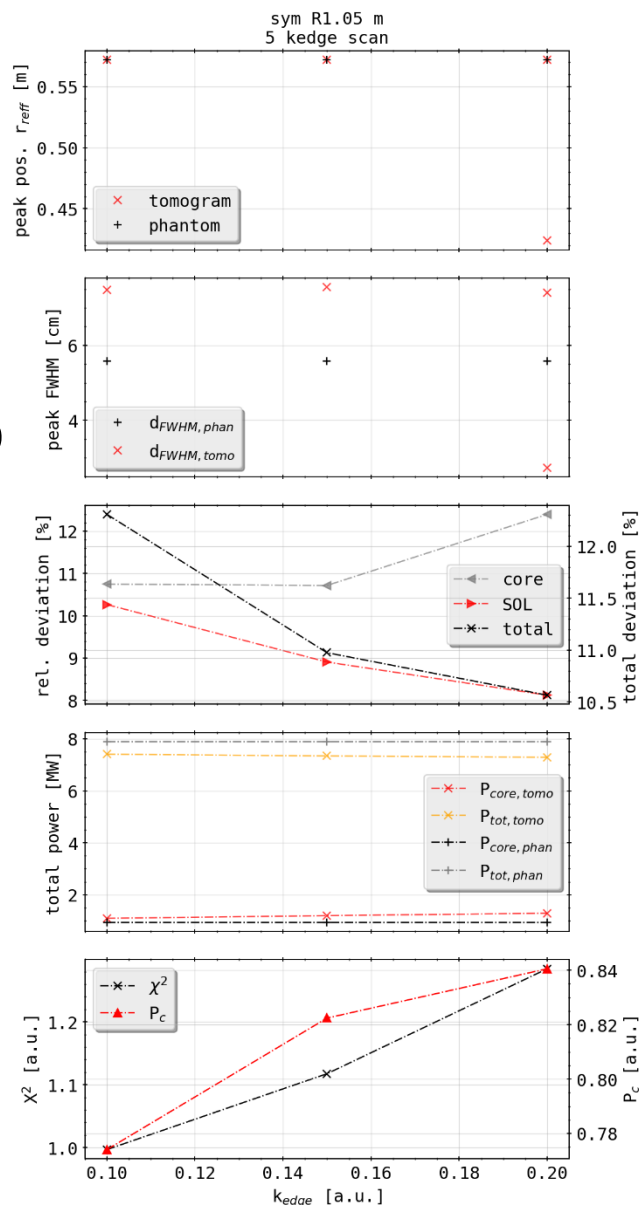


Phantom Test: Island Mimic and Artificial Cameras

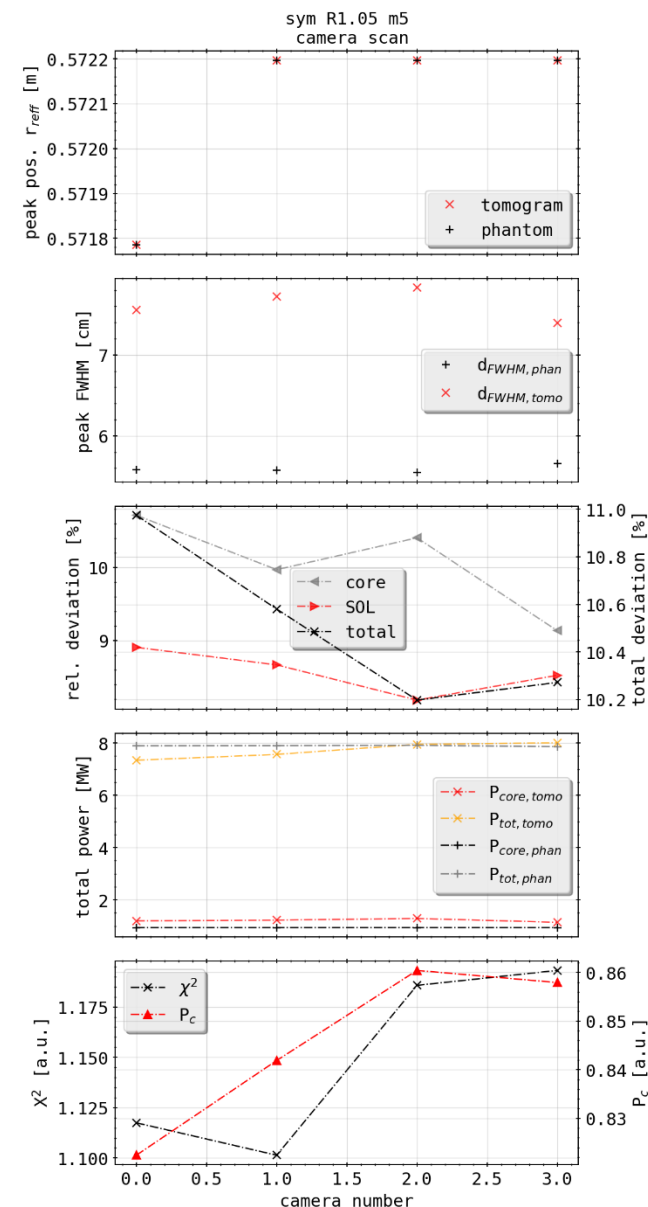
sym R1.05 m5 31x75x4 mx1.0e+06 ARTm aniM
4 1.0 0.15 nT19 nW1 ARTf nigs1 times0.11



K edge scan

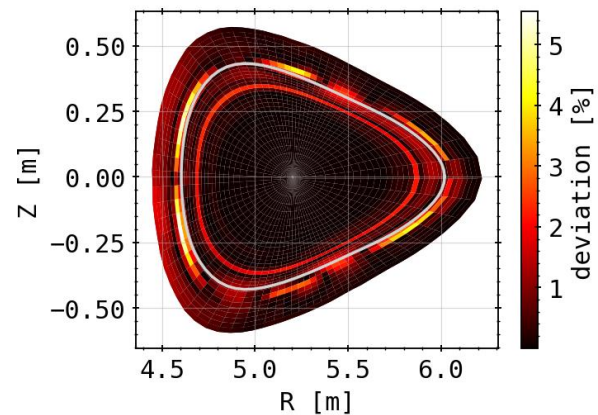
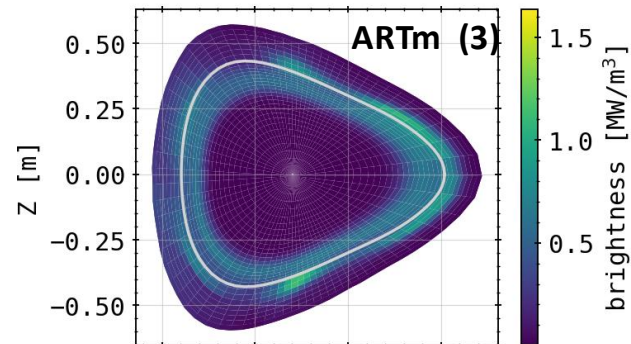
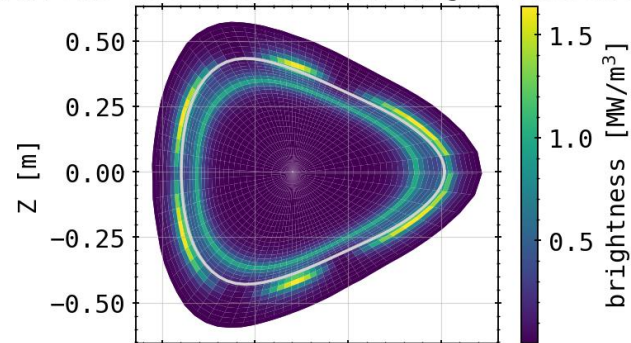


Camera scan

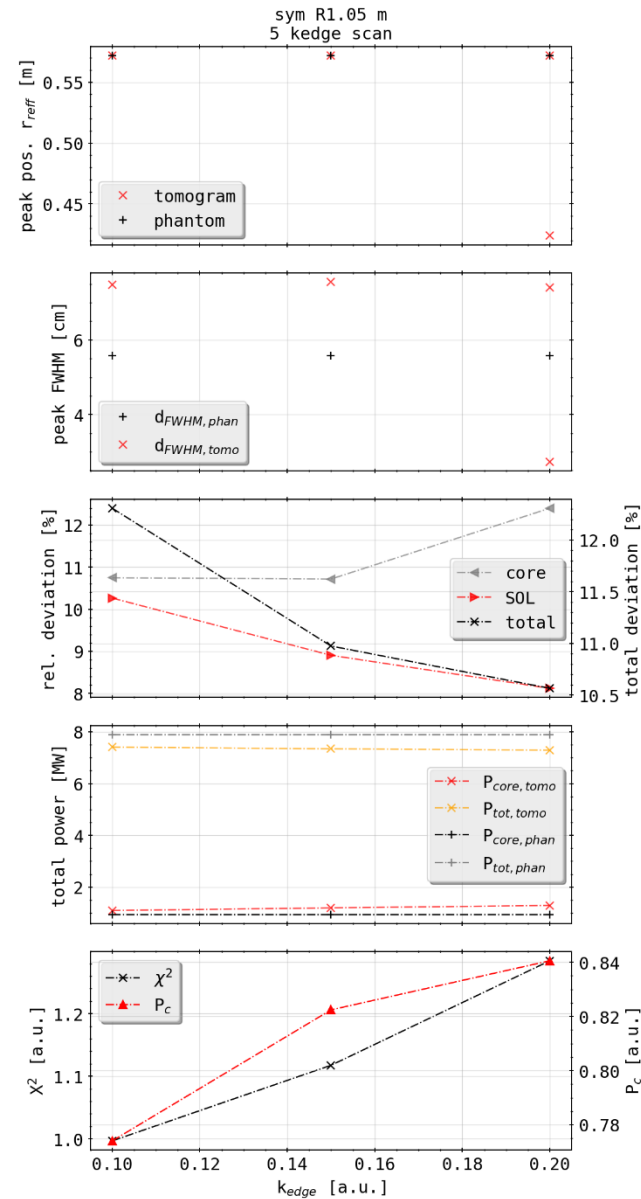


Phantom Test: Island Mimic with Ring and Artificial Cameras

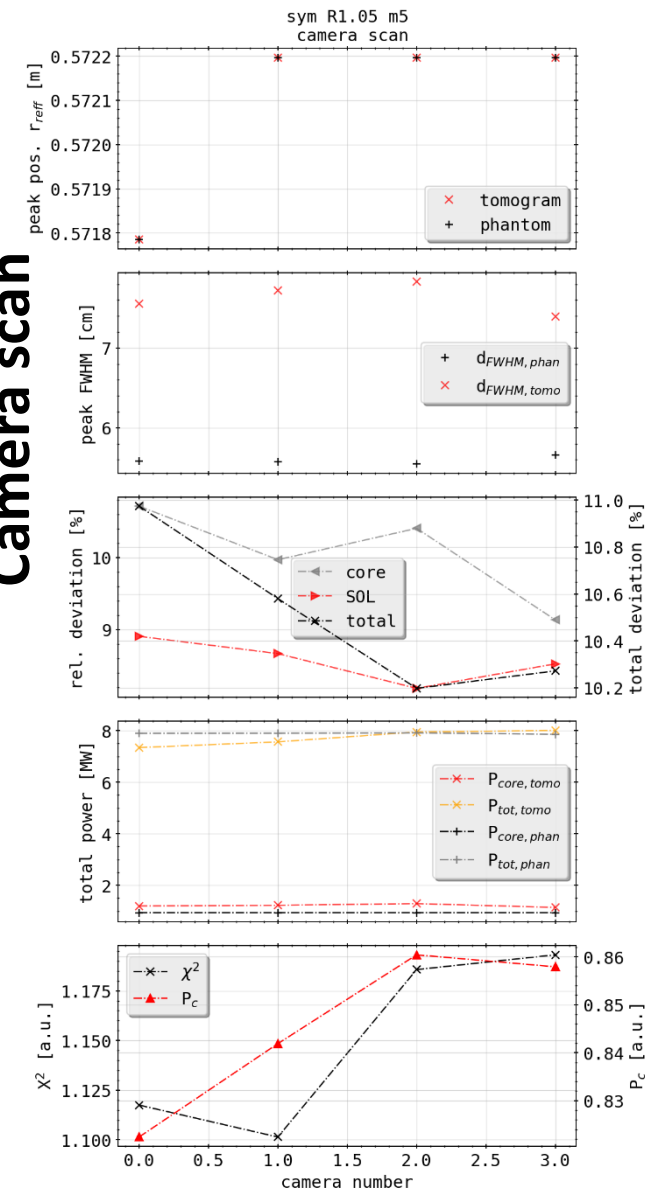
sym R1.05 fsR0.85 31x75x4 mx1.0e+06 ARTm a
niM4 1.0 0.15 nT19 nW1 ARTf nigs1 times0.11



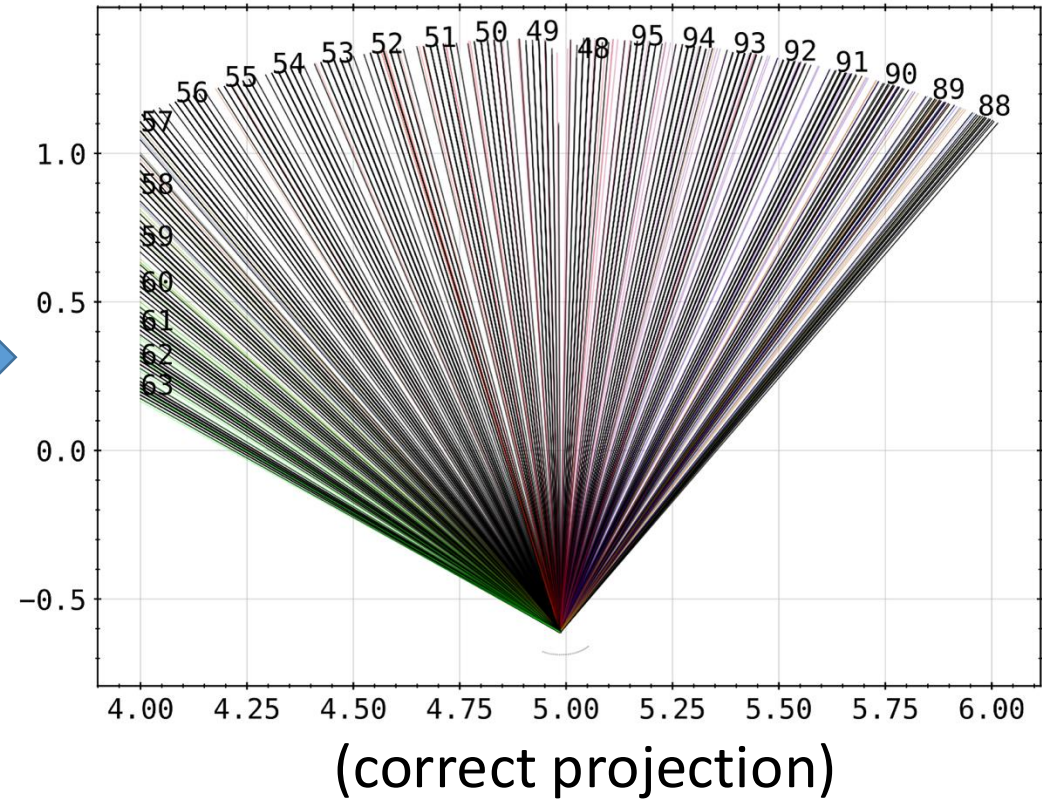
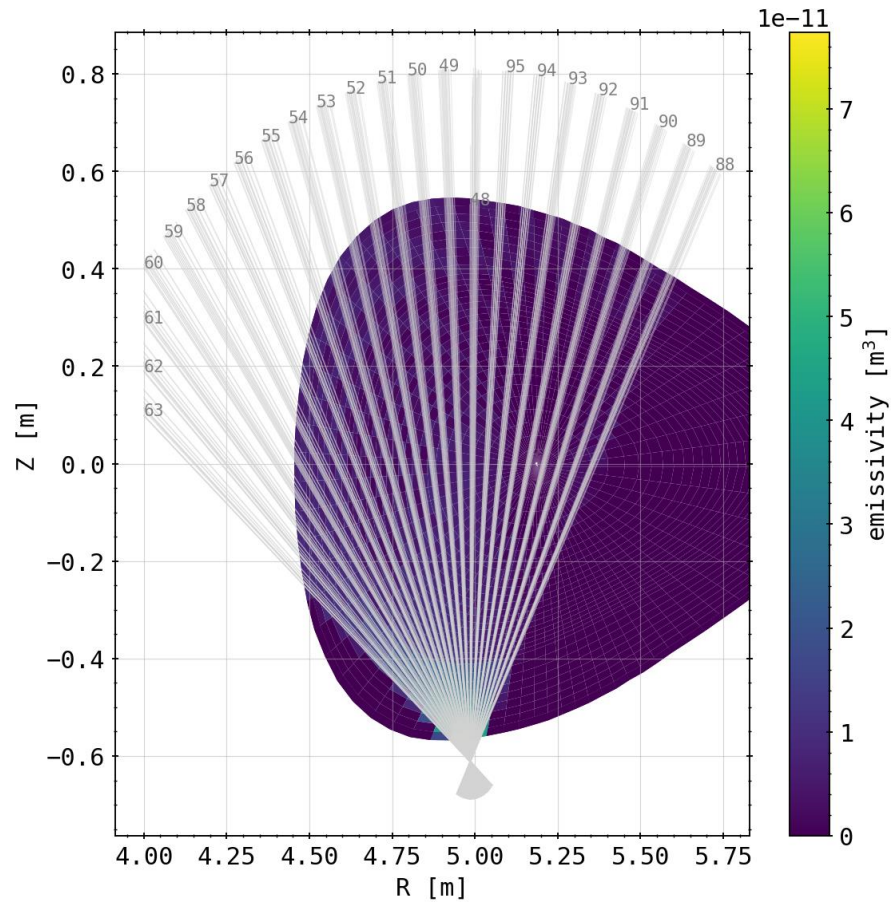
K edge scan



Camera scan

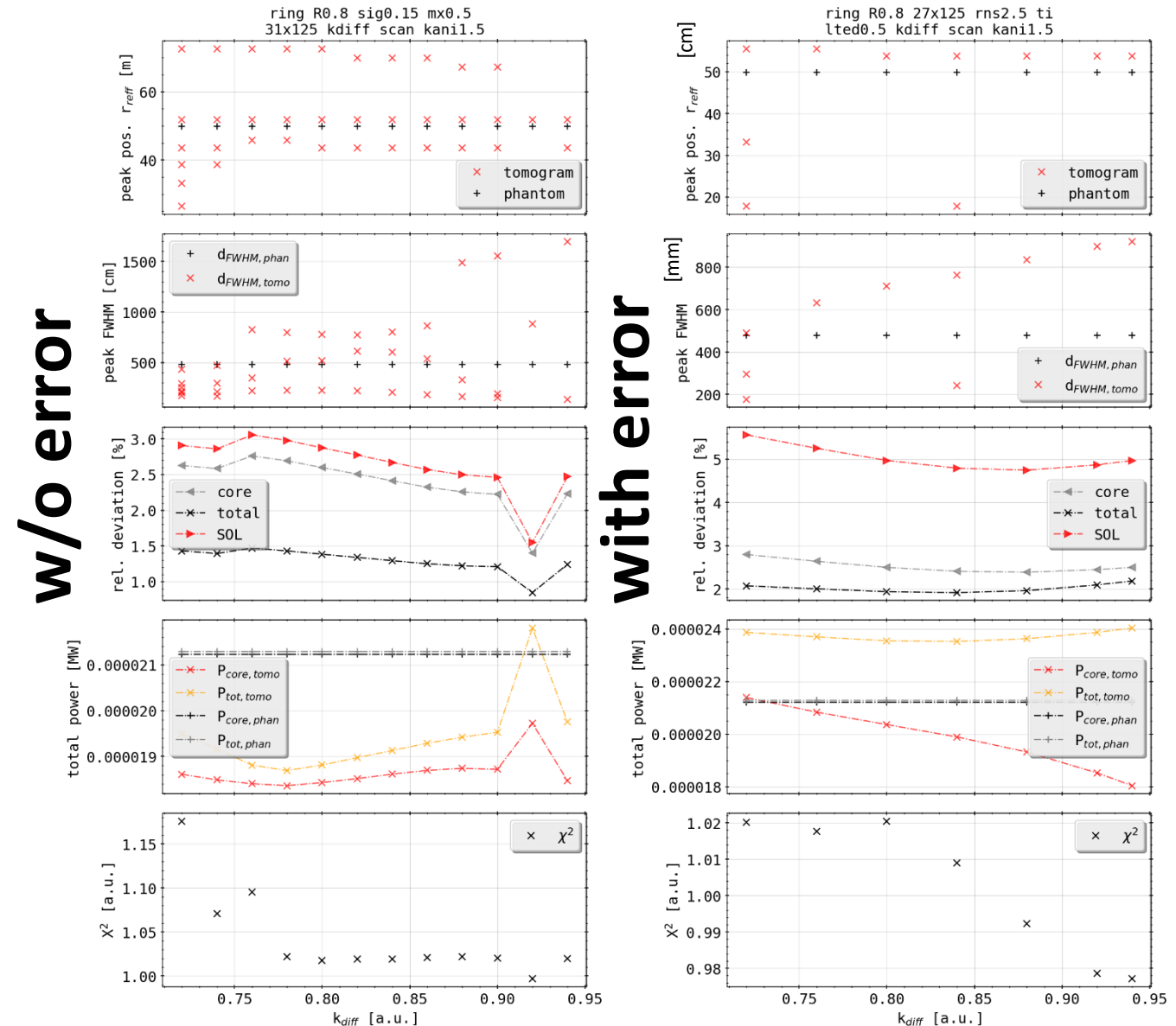


Issues: Line of Sight Geometry

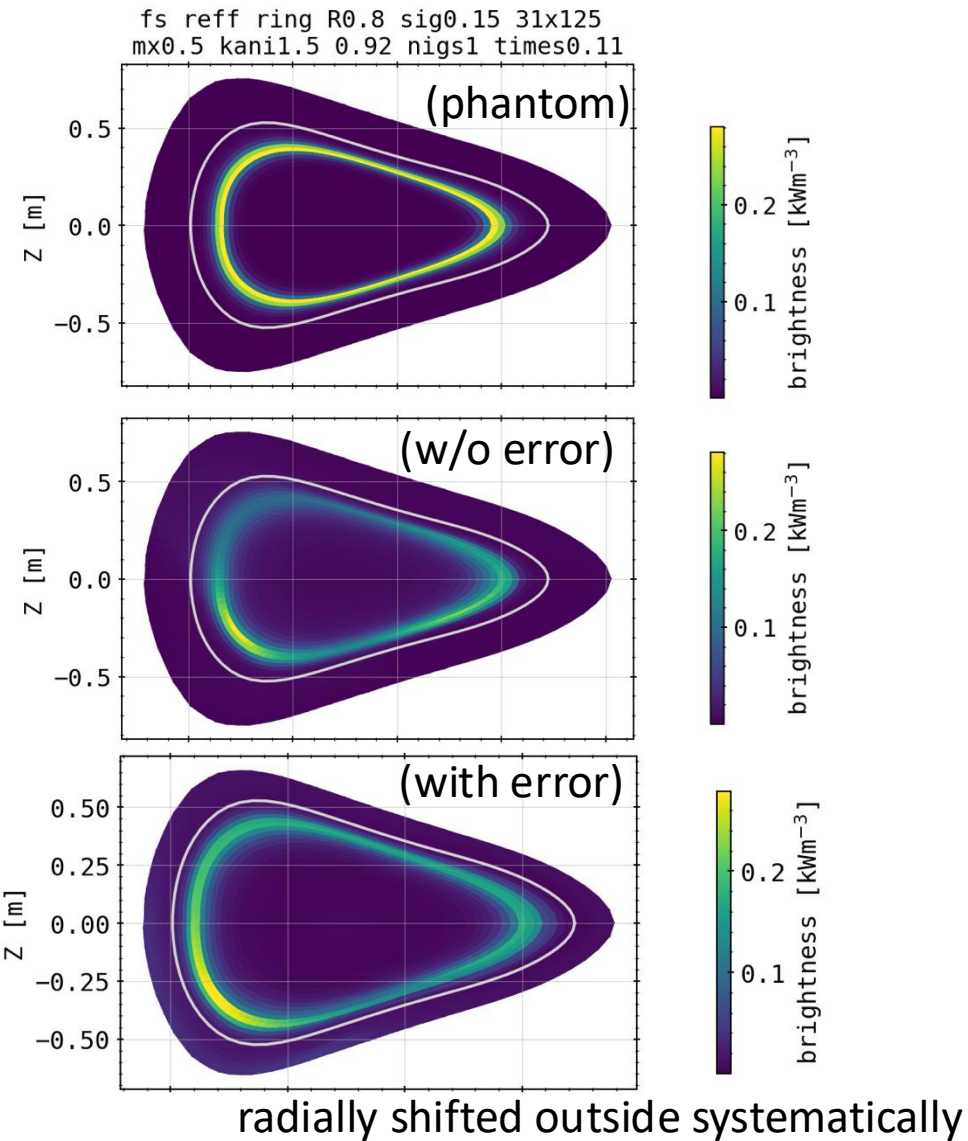


Phantom Test: Ring with 2.5% Random Error and Tilted Camera

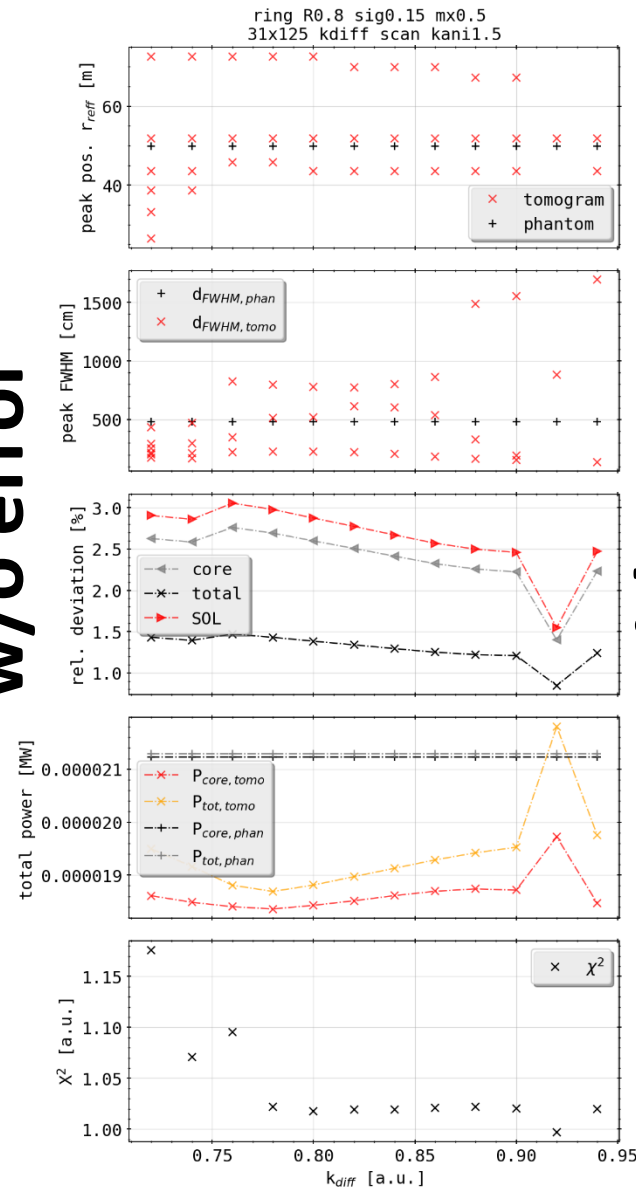
- use geometry previously set up when investigating the effects of errors to emissivities
- add similar 2.5% noise to chordal profile, simulating actually measured experimental profile
- change of best fit parameters according to 2D deviation and χ^2 measurement
- standard error in peak localization radially changed and increased



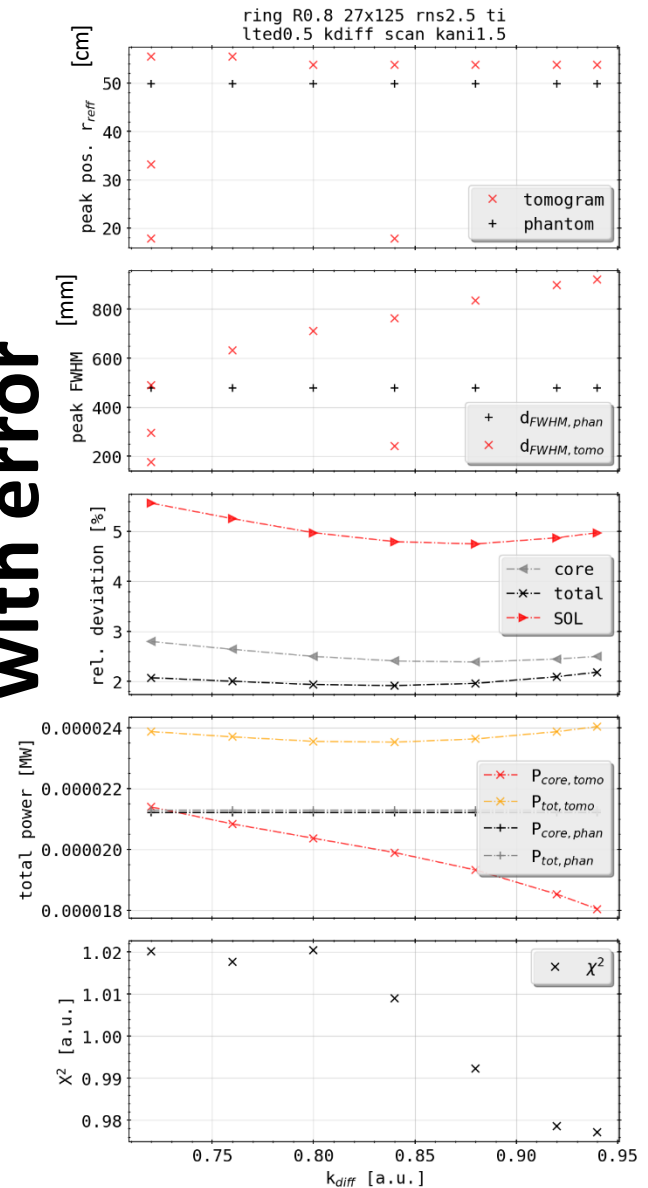
Phantom Test: Ring with 2.5% Random Error and Tilted Camera



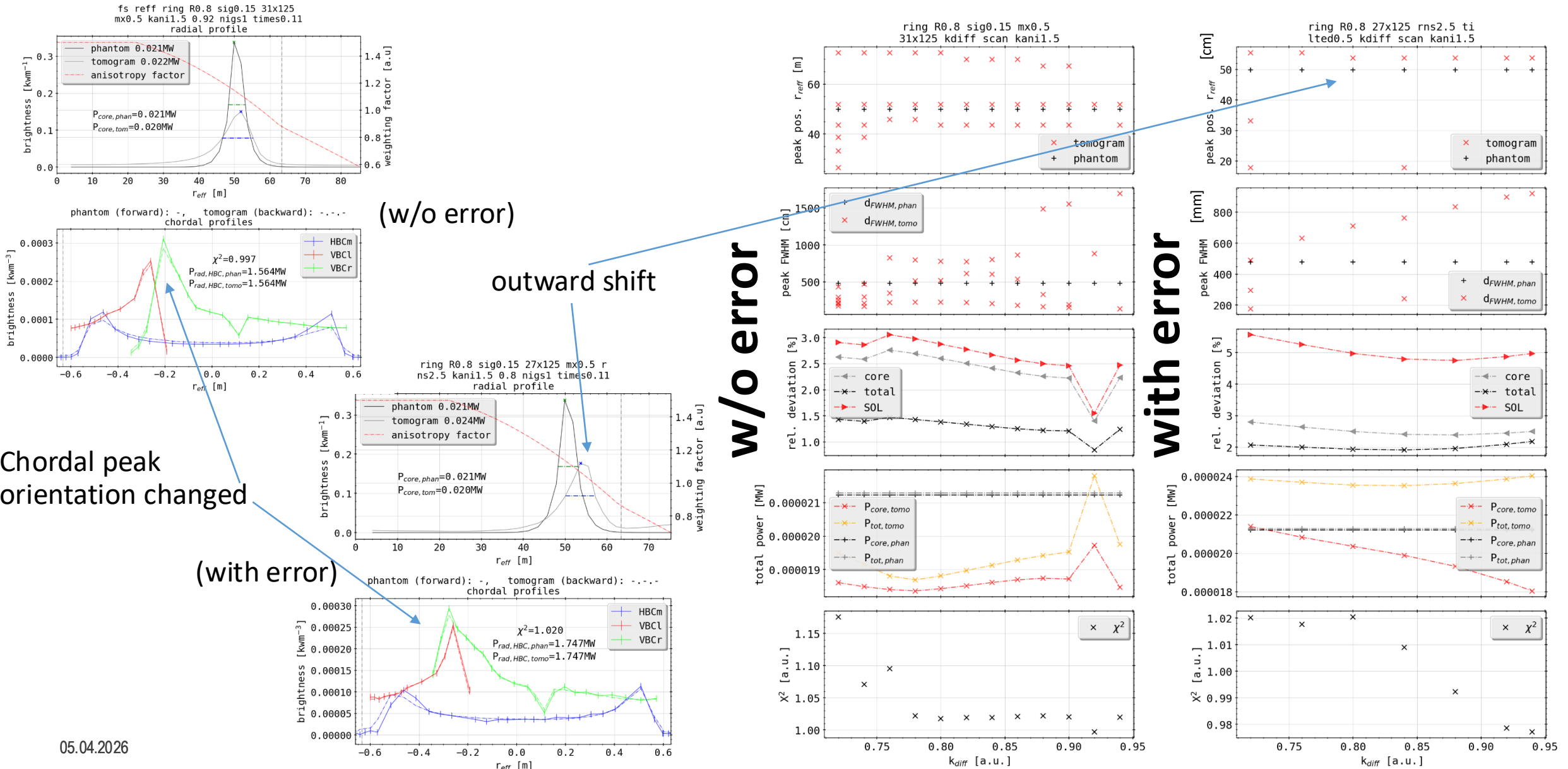
w/o error



with error

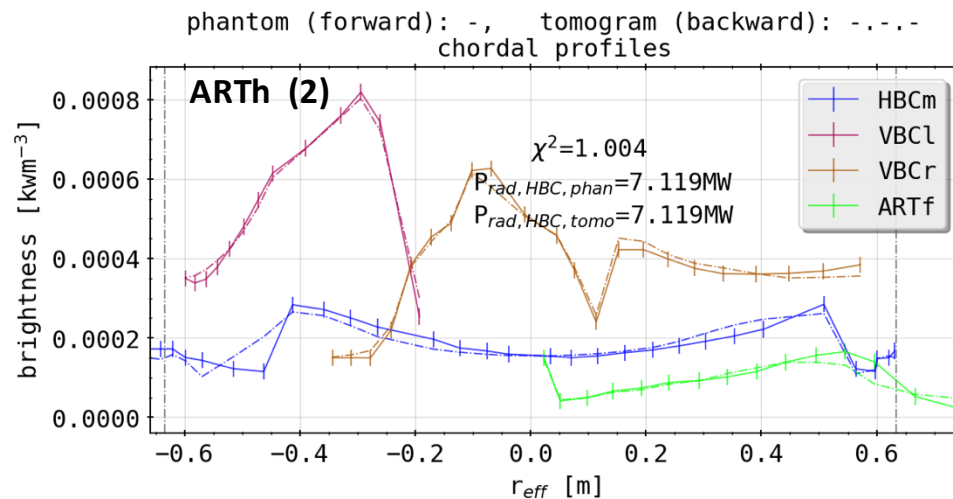


Phantom Test: Ring with 2.5% Random Error and Tilted Camera

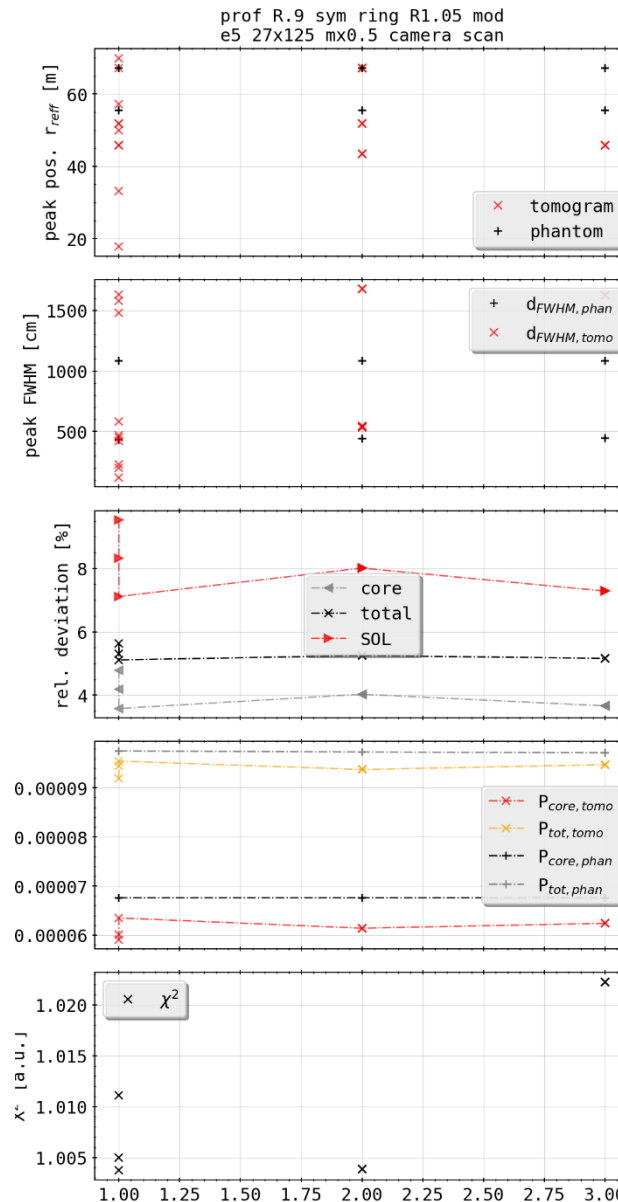


Phantom Test: Island Mimic and Radial Profile

- radial profile inside shadowing over island structure
- virtually not reproducible if poloidally symmetric emissivity profile is superimposed
- outside emissivity focused in front of HBC
- χ^2 fits overall better



amera scan



K factor scan

